

PrepME AI — MongoDB Simplified Notes

1. Why We Needed MongoDB

Before MongoDB:

```
Frontend
  ↓
Backend
  ↓
console.log(req.body)
```

Backend only printed data.

Example:

```
{
  email: "abc@gmail.com",
  password: "123456"
}
```

Problem:

- data disappears after refresh/restart
- nothing is permanently saved

So we needed:

Permanent Storage

That is why databases exist.

2. What MongoDB Actually Is

MongoDB is a database.

Simple meaning:

Online storage room for application data

Backend can:

- save data
- read data
- update data
- delete data

Examples of stored data:

- users
- passwords
- notes
- messages
- progress

3. Current Application Architecture

Frontend
↓
Backend
↓
MongoDB Database

VERY IMPORTANT:

Frontend NEVER directly talks to database.

Backend acts as:

- middle layer
- security layer
- logic handler

4. Important Understanding

At current stage we have ONLY:

Connected backend to database

We have NOT yet:

- stored users
- created collections manually
- fetched data
- updated data
- deleted data

Only connection setup is completed.

5. Real World Analogy

Think like this:

MongoDB

= warehouse

Backend

= warehouse manager

Frontend

= customer

Right now:

- manager got warehouse access
 - but no boxes/data stored yet
-

6. MongoDB Atlas

MongoDB Atlas is cloud-hosted MongoDB service.

Purpose:

- host database online
- avoid local DB setup

- provide free cloud database

Why we used Atlas:

- easier for beginners
 - accessible from anywhere
 - widely used in learning projects
-

7. Database User

Atlas asked for:

- username
- password

Why?

Because backend must authenticate before accessing database.

Similar to login system.

8. Connection String

Example:

```
mongodb+srv://username:password@cluster.mongodb.net/
```

Purpose:

```
Database address + credentials
```

Like website URL, but for database connection.

9. Why .env File Is Used

Database connection string contains secrets.

Examples:

- passwords
- credentials
- API keys

Bad practice:

```
const password = "mypassword"
```

Good practice:

```
MONGO_URI=...
```

This keeps secrets separate from code.

10. dotenv Package

Installed using:

```
npm install dotenv
```

Purpose:

- load .env variables into Node.js

Code:

```
require("dotenv").config()
```

Without this:

```
process.env.MONGO_URI
```

will not work.

11. process.env

Used to access environment variables.

Example:

```
process.env.MONGO_URI
```

Meaning:

```
Read MONGO_URI value from .env file
```

12. Mongoose

Installed using:

```
npm install mongoose
```

Purpose:

```
Bridge between Node.js and MongoDB
```

Mongoose helps:

- connect database
 - create schemas
 - create models
 - perform queries
-

13. mongoose.connect()

Main connection code:

```
mongoose.connect(process.env.MONGO_URI)
```

Meaning:

Backend, connect to MongoDB database using this connection string.

That is the core database connection step.

14. .then()

Example:

```
.then(() => {  
  console.log("MongoDB connected")  
})
```

Purpose:

- runs when connection successful

15. .catch()

Example:

```
.catch((error) => {  
  console.log(error)  
})
```

Purpose:

- handles connection errors

16. URL Encoding Problem

Special characters inside passwords can break connection string.

Examples:

Character	Encoded
@	%40

Character	Encoded
#	%23
&	%26
/	%2F
:	%3A

Example

Actual password:

```
abc@123
```

Connection string version:

```
abc%40123
```

Important:

- actual password stays same
- only URL representation changes

17. Backend Restart Workflow

Whenever backend code changes:

Stop server:

```
CTRL + C
```

Restart server:

```
node index.js
```


Reason:

- Node backend does not auto-refresh by default
-

18. Most Important Understanding

Current stage completed:

Backend ↔ Database Connection

NOT database operations yet.

Actual DB operations will come later:

- save users
- fetch users
- update users
- delete users

That is where MongoDB becomes much easier to understand.

19. Current Full Stack Flow

```
Frontend
  ↓
Axios Request
  ↓
Backend API
  ↓
MongoDB Database
```

Frontend talks to backend.

Backend talks to database.

Frontend never directly accesses DB.

20. Main Learning From MongoDB Setup

The goal was NOT database mastery.

The goal was understanding:

- why databases exist
- how backend connects to database
- why environment variables are used
- how connection strings work
- how modern apps store persistent data